WORKSHEET

Chapter 6: Polynomial operations

1 Given that $p(z) = (5 - 2i)z + 5 + 3i$, $q(z) = 4z^2 + (2 - 7i)z + 5i$ and $r(z) = (4 - 3i)z - 6 + 2i$, calculate and simplify each of the following:

a $p(z) + q(z)$

b $p(z) - q(z)$

c $p(z) + r(z)$

d $r(z) - p(z)$

e $p(z) \times q(z)$

© Cengage Learning Australia Pty Ltd 2014

f $q(z) \times r(z)$

2 Given that $p(z) = (2 + 3i)z^2 + (-7 - 4i)z + 9 + 4i$, $q(z) = (3 + i)z^2 + (5 - 3i)z + 8 - 3i$ and $r(z) = (1 + 3i)z^2 + (2 - 5i)z + 7i$, calculate and simplify each of the following:

a $p(z) + q(z)$

b $p(z) - q(z)$

c $p(z) + r(z)$

d $r(z) - q(z)$

e $q(z) \times r(z)$

f $p(z) \times q(z)$

3 Given that $p(z) = (-2 + 3i)z^2 + (5 - 2i)z + 6 - 3i$, $q(z) = (3 - 4i)z^3 + (7 + i)z^2 + (3 - 2i)z + 10 + 8i$ and $r(z) = (1 - 3i)z^2 + (-5 - i)z + 6 + 4i$, calculate and simplify each of the following:

a $p(z) + q(z)$

© Cengage Learning Australia Pty Ltd 2014

b $p(z) - q(z)$

c $p(z) + r(z)$

d $r(z) - q(z)$

e $q(z) \times r(z)$

© Cengage Learning Australia Pty Ltd 2014

f $p(z) \times q(z)$

4 Given that $p(z) = (1 + 2i)z^3 - (4 + 3i)z^2 + (2 - i)z + 5 - i$, $q(z) = (2 - 3i)z^3 + (5 - i)z^2 + (7 - 3i)z + 5$ and $r(z) = (2 - i)z^3 + (1 - 4i)z^2 - (3 - 5i)z + 2 + i$, calculate and simplify each of the following:

a $p(z) + r(z)$

b $q(z) - p(z)$

c $q(z) + r(z)$

d $p(z) - r(z)$

e $p(z) \times q(z)$

f $q(z) \times r(z)$

5 Given that $p(z) = z^3 + (-5 + 7i)z^2 - (6 + 23i)z + 24 + 18i$, calculate and simplify each of the following:

a $p(z) \div (z - 2)$

b $p(z) \div (z + 3i)$

c $p(z) \div (z - 3 + 4i)$

6 Given that $p(z) = z^3 + (5 + 6i)z - 9 - 10i$, calculate the quotient and remainder for each of the following:

a $p(z) \div (z + 3i)$

b $p(z) \div (z - 2)$

c $p(z) \div (z + 2 - 3i)$

7 CAS Given that $w(z) = (2 - i)z^4 + (3 + 4i)z^3 + (2 - i)z^2 + (6 - 5i)z + 7 - 8i$, $u(z) = (1 - 3i)z^5 + (5 - 3i)z^3 - (2 + 9i)z^2 - 11 + 15i$ and $v(z) = (4 + 5i)z^3 - (-1 + 4i)z - 6 + 5i$, calculate and simplify each of the following:

a $w(z) + u(z)$

b $w(z) - u(z)$

c $w(z) + v(z)$

d $v(z) - u(z)$

e $u(z) \times v(z)$

f $w(z) \times u(z)$

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** $4z^2 + (7 - 9i)z + 5 + 8i$

b $-4z^2 + (3 + 5i)z + 5 - 2i$

c $(9 - 5i)z - 1 + 5i$

d $-(1 + i)z - 11 - i$

e $(20 - 8i)z^3 + (16 - 27i)z^2 + (41 - 4i)z - 15 + 25i$

f $(16 - 12i)z^3 + (-37 - 26i)z^2 + (17 + 66i)z - 10 - 30i$

2 **a** $(5 + 4i)z^2 - (2 + 7i)z + 17 + i$

b $(-1 + 2i)z^2 - (12 + i)z + 1 + 7i$

c $(3 + 6i)z^2 - (5 + 9i)z + 9 + 11i$

d $(-2 + 2i)z^2 - (3 + 2i)z - 8 + 10i$

e $10iz^4 + (25 - i)z^3 + (5 + 11i)z^2 + (22 - 11i)z + 21 + 56i$

f $(3 + 11i)z^4 + (2 - 10i)z^3 + (1 + 40i)z^2 - (11 + 18i)z + 84 + 5i$

3 **a** $(3 - 4i)z^3 + (5 + 4i)z^2 + (8 - 4i)z + 16 + 5i$

b $(-3 + 4i)z^3 + (-9 + 2i)z^2 + 2z - 4 - 11i$

c $-z^2 - 3iz + 12 + i$

d $(-3 + 4i)z^3 + (-6 - 4i)z^2 + (-8 + i)z - 4 - 4i$

e $(-9 - 13i)z^5 - (9 + 3i)z^4 - (3 + 35i)z^3 + (55 + 19i)z^2 - (16 + 50i)z + 28 + 88i$

f $(6 + 17i)z^5 - (10 + 7i)z^4 + (43 - 29i)z^3 + (12 - 17i)z^2 + (78 - i)z + 84 + 18i$

4 **a** $(3 + i)z^3 + (-3 - 7i)z^2 + (-1 + 4i)z + 7 + 4i$

b $(1 - 5i)z^3 + (9 + 2i)z^2 + (5 - 2i)z + i$

c $(4 - 4i)z^3 + (6 - 5i)z^2 + (4 + 2i)z + 7 + i$

d $(-1 + 3i)z^3 + (-5 + i)z^2 + (5 - 6i)z + 3 - 2i$

e $(8 + i)z^6 + (-10 + 15i)z^5 + (-9 - 8i)z^4 + (-16 - 23i)z^3 + (15 - 38i)z^2 + (42 - 27i)z + 25 - 5i$

f $(1 - 8i)z^6 + (-1 - 18i)z^5 + (21 - 15i)z^4 + (2 - 12i)z^3 + (10 + 27i)z^2 + (2 + 26i)z + 10 + 5i$

5 **a** $z^2 + (-3 + 7i)z - 12 - 9i$

b $z^2 + (-5 + 4i)z + 6 - 8i$

c $z^2 + (-2 + 3i)z - 6i$

6 **a** Quotient $= z^2 - 3zi - 4 + 6i$; remainder $= 9 + 2i$

b Quotient $= z^2 + 2z + 9 + 6i$; remainder $= 9 + 2i$

c Quotient $= z^2 + (-2 + 3i)z - 6i$; remainder $= 9 + 2i$

© Cengage Learning Australia Pty Ltd 2014

7 a $(1 - 3i)z^5 + (2 - i)z^4 + (8 + i)z^3 - 10iz^2 + (6 - 5i)z - 4 + 7i$

b $(-1 + 3i)z^5 + (2 - i)z^4 + (-2 + 7i)z^3 + (4 + 8i)z^2 + (6 - 5i)z + 18 - 23i$

c $(2 - i)z^4 + (7 + 9i)z^3 + (2 - i)z^2 + (7 - 9i)z + 1 - 3i$

d $(-1 + 3i)z^5 + (-1 + 8i)z^3 + (2 + 9i)z^2 + (1 - 4i)z + 5 - 10i$

e $(19 - 7i)z^8 + (24 + 6i)z^6 + (46 - 23i)z^5 + (-7 - 23i)z^4 + (-172 + 47i)z^3 + (57 + 44i)z^2 + (49 + 59i)z - 9 - 145i$

f $(-1 - 7i)z^9 + (15 - 5i)z^8 + (6 - 18i)z^7 + (5 - 28i)z^6 + (20 - 75i)z^5 + (-5 - 18i)z^4 + (-139 - 104i)z^3 + (-93 - 6i)z^2 + (9 + 145i)z + 43 + 193i$

© Cengage Learning Australia Pty Ltd 2014